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Caspase-3 Activation by Curcumin in Breast Cancer Cells: Assessment by Flow Cytometry and Acridine Orange/Ethidium Bromide Staining

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ABSTRACT

Background: Curcumin, a bioactive compound from turmeric, has been recognized for its ability to induce apoptosis in various cancer cell lines, including breast cancer. Caspase-3, a critical executioner in the apoptosis pathway, is a key biomarker for evaluating apoptosis. This study aimed to evaluate the effect of curcumin on caspase-3 activation in MCF-7 breast cancer cells, using flow cytometry as the primary analytical method and Acridine Orange/Ethidium Bromide (AO/EB) staining as a confirmatory approach. By employing both techniques, the study sought to validate the observed apoptotic effects of curcumin and demonstrate the potential of AO/EB staining as a practical alternative for apoptosis assessment in laboratories with limited resources.

Methods: Breast cancer cell lines MCF-7 were treated with varying concentrations of curcumin (0, 70, and 80 µg/mL) with 24-hour incubation. Caspase-3 activity was analyzed using flow cytometry and confirmed through AO/EB staining to visualize apoptotic and necrotic cells. Kruskal-Wallis and Mann-Whitney tests were used to assess the significant differences between treatment groups with the degree of toughness $p < 0.05$.

Results: Curcumin significantly ($p < 0.05$) induced apoptosis in MCF-7 breast cancer cell lines, with a dose-dependent increase in caspase-3 activity. AO/EB staining confirmed the flow cytometry results, showing an increase in apoptotic and necrotic cells.

Conclusions: Curcumin increases the activation of ¹caspase-3 in MCF-7 breast cancer cell lines, both flow cytometry and AO/EB staining. The comparison of flow cytometry and AO/EB staining highlights the reliability of AO/EB staining as a viable alternative for apoptosis detection, particularly in research with limited resources. Future research should further elucidate the molecular pathways involved in curcumin-induced apoptosis by *in vivo* studies.

KEYWORDS: AO/EB staining, breast cancer, caspase-3, curcumin, flow cytometry

INTRODUCTION

Caspase-3 is a critical executioner protease in the apoptotic pathway, serving as a central effector in both intrinsic (mitochondrial-mediated) and extrinsic (death receptor-mediated) cell death mechanisms (1–4). Activation of caspase-3 leads to the proteolytic cleavage of key cellular substrates, resulting in the characteristic morphological

and biochemical features of apoptosis (5). Apoptosis, or programmed cell death, plays a fundamental role in maintaining cellular homeostasis by eliminating damaged, senescent, or potentially malignant cells, thereby preventing uncontrolled cell proliferation, which is a hallmark of cancer (6–8). Dysregulation of apoptotic signaling pathways has been implicated in the pathogenesis of numerous diseases, including cancer, autoimmune disorders, and neurodegenerative conditions, underscoring the importance of understanding the molecular mechanisms governing apoptosis (9–11).

Curcumin, a naturally occurring polyphenolic compound derived from *Curcuma longa*, has attracted considerable attention due to its potent anticancer properties and favorable safety profile (12). Extensive experimental evidence demonstrates that curcumin can induce apoptosis in a wide range of cancer cell lines, including breast cancer, through modulation of multiple signaling pathways (13,14). Notably, curcumin has been shown to activate caspase-3 and other apoptotic markers, such as Bax upregulation, Bcl-2 downregulation, cytochrome *c* release, and PARP cleavage, ultimately promoting apoptotic cell death in malignant cells (15–17). In addition to its pro-apoptotic effects, curcumin exhibits anti-inflammatory and antioxidant activities that further contribute to its therapeutic potential in oncology (18).

Flow cytometry is a widely used and highly sensitive technique for the quantitative analysis of apoptosis, allowing precise detection of apoptotic markers such as activated caspase-3 through fluorescence-labeled antibodies or substrates (19,20). This method enables objective and reproducible measurement of apoptotic responses following exposure to anticancer agents, including curcumin (25). In parallel, the dual staining method using acridine orange (AO) and ethidium bromide (EB) remains a practical and well-established approach for distinguishing viable, apoptotic, and necrotic cells based on membrane integrity and nuclear morphology (21). AO permeates both viable and non-viable cells, emitting green fluorescence when bound to DNA, whereas EB selectively enters cells with compromised membranes and emits orange to red fluorescence upon intercalation with nucleic acids (22).

Using AO/EB staining, cells can be morphologically classified into distinct states: viable cells exhibiting uniform green nuclei, early apoptotic cells showing condensed or fragmented green nuclei, late apoptotic cells displaying orange or red fragmented nuclei, and necrotic cells characterized by uniformly orange or red nuclei (23,24). Despite its advantages, EB is known to be toxic and mutagenic, necessitating strict handling procedures and limiting its application in *in vivo* studies (25). Consequently, in the present study, AO/EB staining was applied primarily to distinguish living and dead cells based on membrane integrity, where green fluorescence indicated viable cells and orange or yellowish fluorescence indicated non-viable cells.

While flow cytometry provides superior quantitative accuracy and sensitivity for apoptosis detection, AO/EB staining offers a rapid, cost-effective, and visually interpretable alternative that does not require sophisticated instrumentation (26,27). Therefore, this study was conducted to evaluate the effect of curcumin on caspase-3 activation in MCF-7 breast cancer cells using flow cytometry as the primary quantitative method, with AO/EB staining employed as a complementary technique to confirm apoptotic morphology. This dual-method approach aims to demonstrate that AO/EB staining can reliably support flow cytometric findings and serve as a feasible alternative for apoptosis assessment in laboratories with limited access to advanced cytometry equipment. We hypothesize that curcumin induces apoptosis in MCF-7 cells through caspase-3 activation and that apoptotic changes detected by flow cytometry can be consistently confirmed using AO/EB staining.

The novelty of this study lies in the integrated application of flow cytometry and AO/EB staining to evaluate curcumin-induced caspase-3 activation in breast cancer cells. By validating a simple staining technique against a gold-standard quantitative method, this research expands the practicality of apoptosis detection in resource-constrained settings while contributing to a deeper understanding of the pro-apoptotic mechanisms of curcumin.

METHODS AND MATERIALS

This research was designed as an experimental laboratory study conducted from January to March 2024 at the integrated laboratory, Faculty of Medicine, University of YARSI, Jakarta, Indonesia, and the integrated laboratory, Faculty of Medicine, University of Indonesia, Jakarta, Indonesia. A total of nine experimental samples were analyzed, consisting of three curcumin concentration groups (0, 70, and 80 µg/mL) with three independent replicates per group.

The MTS kit was obtained from the Promega (Madison, USA). The caspase-3 activity was determined using PE Active Caspase-3 Apoptosis Kit, BD Pharmingen™, BD Biosciences

CELL CULTURE

Breast cancer cell lines MCF-7 (ATCC® HTB-22™) were purchased from ATCC (Manassas, VA, USA) at the second passage. Breast cancer MCF-7 cells were routinely cultured in tissue flask containing Dubelco's minimum essential medium (DMEM) (Gibco, New York, USA) with 10% fetal bovine serum (FBS from Biowest, Riverside, USA), 5% penicillin/streptomycin, and 5% amphotericin (Biosciences, San Jose, USA) at 37°C in a humidified atmosphere with 5% CO₂. Cell culture media were changed every 2 days in a sterile manner. To determine the effects of curcumin on cell proliferation, cells from stock plates were harvested with 0.25% trypsin, buffered with 0.2% EDTA (pH 7.3), and counted using a haemocytometer. Cells were adjusted to a density of 1x10⁶ cells/well to the required volume of plating medium supplemented with 10% PBS and curcumin was dissolved in dimethyl sulfoxide (DMSO) before use.

SAMPLE PREPARATION

Curcumin was purchased from Sigma Chemical Co. (St. Louis, MO, USA). Curcumin is dissolved in DMSO. In the well plate 96, the total number of cells 5x10³/well is inserted. The sample was divided into 3 groups: control (0 µg/mL), 70 µg/mL, and 80 µg/mL of curcumin. Then, a solution of curcumin concentration (0, 70, and 80 µg/mL) and a complete culture medium was added so that the final volume of each concentration series for treatment was 100 µL/well. The cells in the well plate 96 were stored in an incubator at 37°C with a humidity of 5% CO₂ for 24 hours.

CYTOTOXICITY (IC₅₀) ASSAY

Previously, the IC₅₀ of curcumin was tested on MCF-7 cells using the MTS kit. MTS and PMS were removed 90 minutes before use. The required solution mixture was calculated using the following formula:
$$MTS = \{(\text{number of wells to be treated} + 2) \times 20 \mu\text{L}\} + PMS \{(\text{number of wells to be treated} \times 2)\}$$

To cultivate cells in each well of a 96-well plate, the total number of cells required was 5 x 10³/well. Then, 10 µL of curcumin concentration solution and complete culture medium were added, bringing the final volume of each treatment concentration series to 100 µL/well. Blank wells (complete culture medium without cells and without curcumin) and cells without curcumin were also added as negative controls. The cells in the 96-well plate were kept in an incubator at 37°C with 5% CO₂ humidity for 48 hours. Next, 20 µL of MTS and PMS solution were added to the biosafety cabinet (BSC) with the lights turned off. After being wrapped in aluminum foil, the 96-well plate was placed back in an incubator at 37°C with 5% CO₂ humidity for 2 hours. The optical density was then determined using a microplate reader set to 490 nm.

CASPASE-3 EXAMINATION

Flow cytometry

The caspase-3 activity was determined using flow cytometry according to the manufacturer's instructions (catalog No. 550914 from BD PharMingen). Cells were washed twice with ice-cold PBS, then resuspended in Cytofix/Cytoperm™ solution at a concentration of 1x10⁶ cells/0.5 mL. The cells were incubated for 20 minutes on ice. The cell pellet, aspirate, and Cytofix/Cytoperm™ solution were washed twice with Perm/Wash™ buffer at a volume of 0.5 mL buffer/1x10⁶ cells at room temperature. The cells, Perm/Wash™ buffer, and antibody were resuspended and incubated for 30 minutes at room temperature. Each test was washed in 1 mL of Perm/Wash™ buffer, then resuspended in 0.5 mL of Perm/Wash™ buffer. The analysis was then performed using a flow cytometer. Cisplatin was used as the positive control and the untreated sample was used as the negative control. Fold-increase in caspase activity was determined by comparing the results of treated samples with the level of the untreated control. Calculation of the number of caspase3 using the formula: The number of cells showing an autofluorescent pattern is divided by the total number of cells multiplied by 100%.

Acridine Orange/Ethidium Bromide (AO/EB) Staining

Cells of 1x10⁶ cells/mL were washed 2 times with cold phosphate buffer. A double staining solution (10 µL) containing 100 µg/mL AO and 100 µg/mL EB is added to each suspension. Incubate the cells for 5 minutes. Then

the cell suspension (10 µL) is transferred to the slide glass and covered with a lid. The double staining method of AO/EB was repeated 3 times. Cell morphology is examined and 100 cells are counted using a fluorescent microscope. The assessment of AO/EB staining was divided into two groups: living cells and dead cells (apoptosis and necrotic) by observing chromatin condensation or nuclear fragmentation and changes in fluorescence colour. Cell morphology was read by two observers in a double-blind manner. Acridin orange/ethidium bromide staining was carried out as confirmation of cytotoxicity and caspase-3 activity by assessing changes in cell morphology using a fluorescent microscope. Living cells are green and dead cells are orange or yellowish. The calculation uses the AO/EB staining method by assessing cell death (apoptotic and necrotic) because it is difficult to assess between late apoptotic cells and necrotic cells because the color of the cells is similar, namely orange/red nuclei. Calculation of the number of dead cells using the formula: Percentage of dead cells = number of dead cells divided by 100 cells multiplied by 100%

⁴ The average growth rate (delta) is calculated using the formula:

$$\text{Growth rate (delta)} = \frac{(\text{post exposure data} - \text{without exposure data})}{\text{without exposure data}} \times 100\% \quad (1)$$

STATISTICAL ANALYSIS

Data are expressed as mean±standard deviation (SD) from at least three independent experiments. The results of the normality test of caspase-3 examination using flow cytometry and AO/EB staining showed that the data distribution was not normal, so the Kruskal-Wallis statistical test was performed, followed by the Mann-Whitney Post hoc test for differences among multiple groups. The correlation test between flow cytometry and AO/EB staining used the Spearman non-parametric test. Significance was assigned for p values < 0.05 using the SPSS 20.0 statistical software for Windows.

RESULT

Cytotoxicity (IC50) Assay

The IC50 results for curcumin in MCF-7 cells were 75.73 µg/mL, so the curcumin concentrations to be used for this study were 70 µg/mL and 80 µg/mL.

² Caspase-3 examination using Flow Cytometry and Acridine Orange/Ethidium Bromide (AO/EB) Staining

The percentage of caspase-3 activity was increased significantly in both breast cancer cells after curcumin exposure compared to controls. The results of the Kruskal-Wallis test showed a result of $p=0.011$ for caspase-3 examination using flow cytometry and $p<0.001$ for caspase-3 examination using AO/EB staining (Table 1).

Table 1. Caspase-3 Activity by ² Flow Cytometry and Acridine Orange/Ethidium Bromide Staining Based on Curcumin Concentration Group Given to MCF-7 Breast Cancer Cells

Examination	n	Curcumin	Median (minimum–maximum)	Delta (%) Compare to control	p
Flow Cytometry	3	No Exposure	1,12 (0,38–2,25)	0	< 0,001
	3	Low Concentration (70µg/mL)	71,94 (53,19–83,29)	6323,21	
	3	High Concentration (80µg/mL)	94,75 (87,23–96,55)	8359,82	

AO/EB Staining	3	No Exposure	30 (25–48)	0	0,011
	3	Low Concentration (70µg/mL)	40 (18–91)	285,27	
	3	High Concentration (80µg/mL)	72 (62–78)	448,06	

Kruskal-Wallis Test. Post-hoc Mann-Whitney Test.

Flow Cytometry:

No Exposure vs. Low Concentration $p < 0,001$; No Exposure vs. High Concentration $p < 0,001$; Low Concentration vs. High Concentration $p < 0,001$.

AO/EB Staining:

No Exposure vs. Low Concentration $p = 0,687$; No Exposure vs. High Concentration $p < 0,001$; Low Concentration vs. High Concentration $p = 0,227$.

The correlation test of caspase-3 examination using flow cytometry and AO/EB staining showed that there was a strong correlation ($p=0.02$, $r=0.750$). Correlation was performed on all concentration groups.

Table 2. Relationship between Caspase-3 Activity using Flow Cytometry and Acridine Orange/Ethidium Bromide Staining in MCF-7 Breast Cancer Cells

Cell Culture	n	correlation coefficient (r)	p
MCF-7	3	0,750	0,020

Spearman correlation test

The results of the analysis showed a meaningful unidirectional relationship between the percentage of caspase-3 activity and AO/EB staining in MCF-7 cells with moderate to strong positive correlation strength. The higher the percentage of caspase-3 activity, the higher the percentage of AO/EB staining.

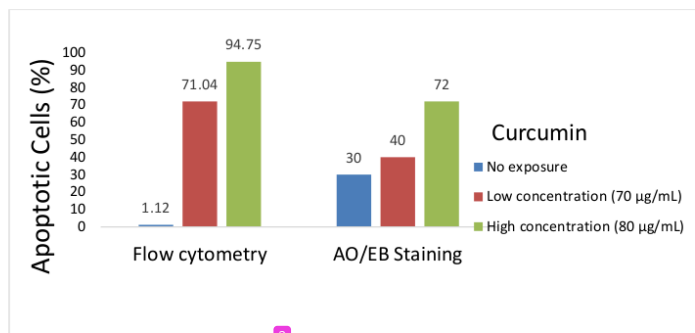


Figure 1. Caspase-3 Activity with Flow cytometry and Acridine Orange/Ethidium Bromide staining Based on Curcumin Concentration Groups Given to Breast Cancer Cells.

The **Figure 1** showed an increase in caspase-3 activity in both breast cancer cells that received curcumin was significant both in flow cytometry and AO/EB staining. The increase in the percentage of caspase-3 activity in both cell lines was highest in the administration of curcumin 80 µg/mL.

Cell Morphology on Caspase-3 Examination using Acridine Orange/Ethidium Bromide (AO/EB) Staining

The results of the examination with fluorescence microscopy were calculated based on 3 staining displays, namely AO staining, EB staining, and AO/EB staining in MCF-7 breast cancer cells (**Figure 2**).

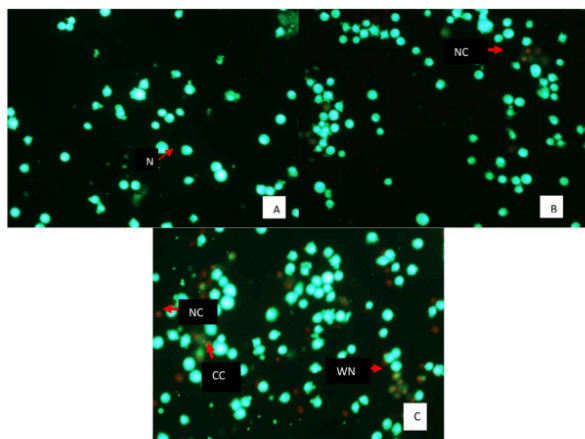


Figure 2. Apoptosis State in MCF-7 Cells with AO/EB Staining (Fluorescence Microscopy, 200x). (A). Without treatment; (B). Curcumin treatment 70 $\mu\text{g/mL}$; (C). Curcumin treatment 80 $\mu\text{g/mL}$ (Arrow N. Normal cells are green; WN. Cells with a wrinkled nucleus are greenish-orange; CC. Cells with a nucleus undergo chromatin condensation; NC. Necrotic cells are characterized by reddish-orange warm nuclei).

DISCUSSION

The results of this study demonstrate that curcumin significantly induces apoptosis in MCF-7 breast cancer cells, as evidenced by increased caspase-3 activity detected by AO/EB staining and quantitatively confirmed by flow cytometry (16,17). Caspase-3 is a central executioner protease in the apoptotic cascade, and its activation is a widely accepted indicator of apoptosis induction in cancer cells (1,28). The observed increase in caspase-3 activity across escalating curcumin concentrations suggests that curcumin effectively promotes programmed cell death rather than nonspecific cytotoxicity. These findings are consistent with previous reports describing the pro-apoptotic effects of curcumin in multiple cancer cell lines, including breast cancer, thereby reinforcing its potential as a therapeutic agent in oncology (15,29).

A key finding of this study is the dose-dependent increase in caspase-3 activity following curcumin exposure, with the highest activation observed at 80 $\mu\text{g/mL}$. This observation aligns with studies demonstrating that curcumin activates apoptosis primarily through the intrinsic mitochondrial pathway (4,7,24). Mechanistically, curcumin has been shown to facilitate mitochondrial membrane permeabilization, resulting in cytochrome-c release into the cytosol and subsequent activation of caspase-9, which in turn activates caspase-3 (8,17). Activated caspase-3 mediates the cleavage of essential cellular substrates, including cytoskeletal proteins, protein kinases, DNA repair enzymes, and endonuclease inhibitors, ultimately leading to chromatin condensation, DNA fragmentation, and apoptotic morphological changes (3,30).

Flow cytometry analysis corroborated the AO/EB staining findings, showing a significant increase in activated caspase-3 in curcumin-treated cells compared with untreated controls. Flow cytometry is widely recognized as a highly sensitive and quantitative method for detecting apoptosis through fluorescent labeling of activated caspases (19,20). In contrast, AO/EB staining provides a simpler and rapid method for distinguishing live, apoptotic, and necrotic cells based on membrane integrity and nuclear morphology (21). The strong agreement between flow cytometry and AO/EB staining observed in this study underscores the reliability of both techniques in detecting

curcumin-induced apoptosis. Furthermore, morphological evaluation via AO/EB staining suggests that lower concentrations of curcumin predominantly induce apoptosis, whereas higher concentrations may also trigger necrotic features, a phenomenon that has been reported in dose-dependent cytotoxic studies (26).

The significant concordance between flow cytometry and AO/EB staining indicates that AO/EB staining can serve as a reliable alternative for apoptosis assessment, particularly in laboratories with limited access to advanced cytometry equipment. Although ethidium bromide is toxic and mutagenic and therefore unsuitable for in vivo applications, its controlled use in in vitro assays remains acceptable with appropriate safety measures (31). The complementary application of flow cytometry and AO/EB staining strengthens the conclusion that curcumin induces apoptosis in MCF-7 cells primarily via caspase-3 activation. Nevertheless, this study has limitations. The interpretation of flow cytometry data focused on apoptotic detection rather than necrosis, while AO/EB staining provided morphological confirmation without detailed mechanistic discrimination. Additionally, the relatively small sample size and the exclusive use of a single breast cancer cell line may limit the generalizability of the findings (32).

Future studies should incorporate a broader range of breast cancer subtypes and larger sample sizes to validate the observed effects of curcumin across heterogeneous tumor models. Moreover, the present study evaluated only short-term curcumin exposure; therefore, long-term investigations are required to determine whether its pro-apoptotic effects are sustained and whether resistance mechanisms emerge over time (12,33). Importantly, in vivo studies are essential to assess the therapeutic relevance of curcumin in the context of tumor microenvironment complexity and systemic pharmacokinetics (6). Clinically, curcumin's low toxicity profile and ability to enhance apoptosis suggest its potential use as an adjuvant agent in breast cancer therapy. Several recent studies have demonstrated that curcumin can sensitize cancer cells to conventional chemotherapeutic agents and reduce drug resistance, highlighting its promise in combination treatment strategies (34,35).

CONCLUSION

Curcumin induced the activation of caspase-3 in MCF-7 breast cancer cell lines, both using flow cytometry and AO/EB staining. The comparison of flow cytometry and AO/EB staining highlights the reliability of AO/EB staining as a viable alternative for apoptosis detection, particularly in research settings with limited resources.

DECLARATIONS

Competing interest

The author(s) declare no competing interest in this study.

Ethics approval and consent to participate

Applicable.

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